

RAM CHARAN VANGA

RA2511027010164

Subject : DSA

Unit 3, S3, SLO2 - Stack Applications - Matching

Match the Stack Applications with their corresponding functional description:

- A. Expression Evaluation**
- B. Reverse Characters**
- C. Syntax Parsing**
- D. Backtracking**
- E. Undo Mechanism**
- F. Function Call Management**
- G. Parenthesis Checking**
- H. Browser History**
- I. Reversing a String**
- J. Tower of Hanoi**

Answer: A - 8 (Converts infix to postfix or prefix and evaluates expressions)

B - 3 (Characters are popped in reverse order)

C - 9 (Helps in checking for balanced symbols / grammar verification)

D - 6 (Used in solving mazes or puzzles via recursive exploration)

E - 7 (Maintains previous states of user actions for rollbacks)

F - 4 (Uses call stack to manage function activation records)

G - 5 (Verifies matching of open and close brackets)

H - 1 (Navigates forward and backward between visited web pages)

I - 10 (Uses stack LIFO property to reverse sequence of characters)

J - 2 (Uses recursion implemented via call stack to transfer disks)